Network address Translation :-
(NAT)

→ The idea of NAT is to allow multiple devices to access the internet through a single public address.

→ To achieve this, the translation of private IP address to a public IP address is required.

→ Network Address Translation (NAT) is a process in which one (or) more local IP address is translated into one or more Global IP address and vice versa in order to provide Internet access to local hosts.

→ Also it does the translation of port numbers, ~~the packet that~~ i.e. masks the ports numbers of the host with another port number, in the packet that will routed to destination.

→ It then masks the corresponding entries of IP address and port number in the NAT table.

→ NAT generally operates on a router or firewall.

→ To seperate the addresses used inside the home or business and the ones used for the Internet, the Internet authorities have reserved 3 sets of address as private addresses.

	Range		Total
10.0.0.0	to	10.255.255.255	2^{24}
172.16.0.0	to	173.31.255.255	2^{20}
192.168.0.0	to	192.168.255.255	2^{16}

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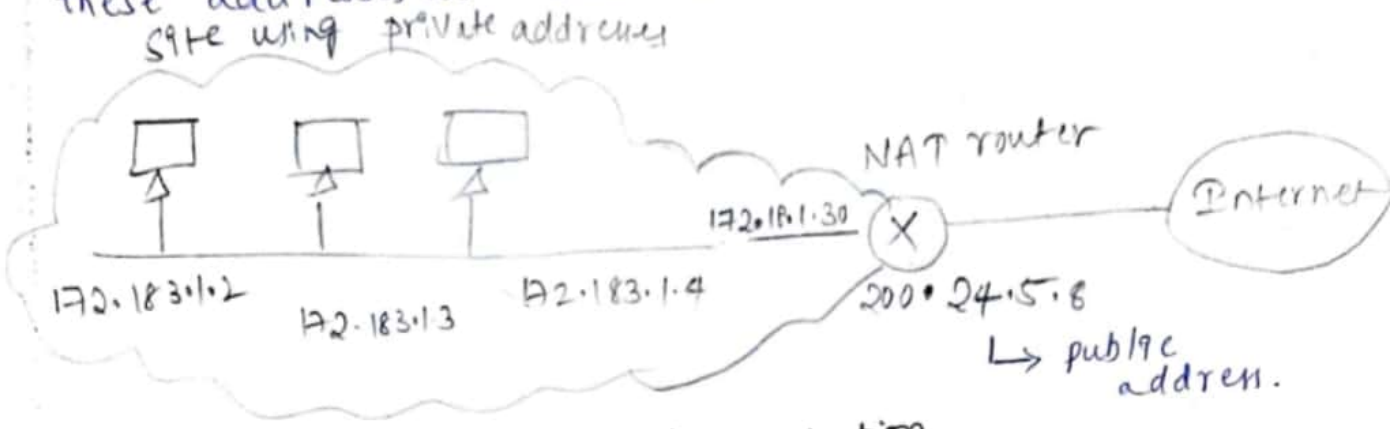
→ Any organization can use an address out of this without permission from the Internet authorities.

→ These are unique inside the organizations, but they are not unique globally.

→ No router will forward a packet that has one of these addresses as destination address.

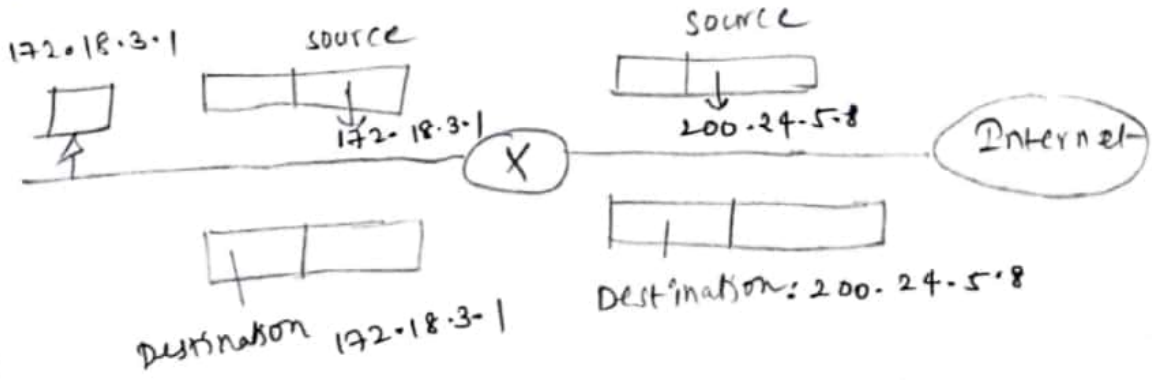


A NAT Implementation

Address Translation :-

→ All the outgoing packets go through the NAT router, which replaces the source address in the packet with global NAT address.

→ All the incoming packets also pass through the NAT router, which replaces the destination address in the packet with appropriate private address.



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ansition Table :

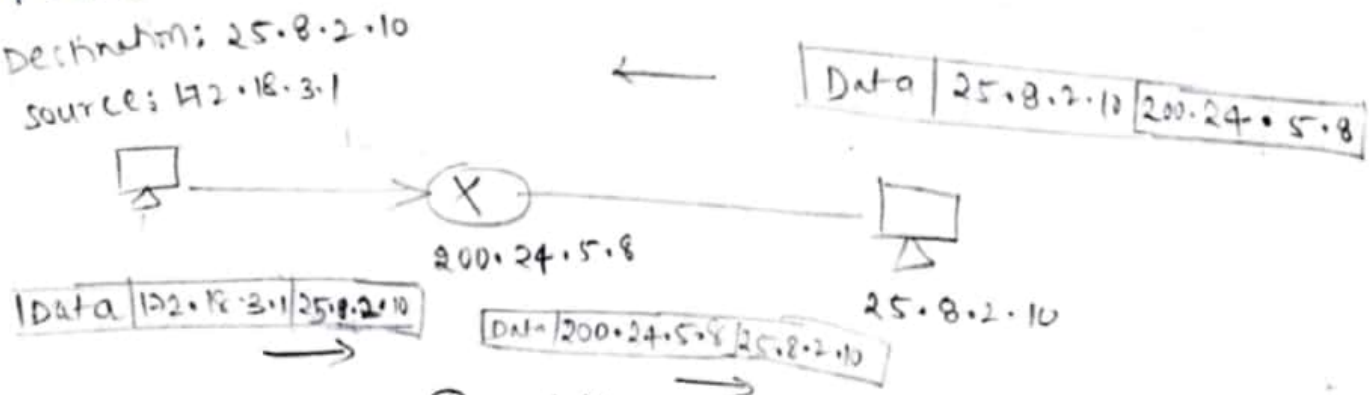
→ Translating the source address for outgoing packet is straightforward.

→ NAT maintains a translation table has only two columns :

Private ip address ↓ source ip address	External address ↓ Destination address.

→ when the router translates the source address of the outgoing packet, it also makes note of the destination address – where the packet is going.

→ When the response comes back from the destination, the router uses the source address of the packet (as external address) to find the private address of the packet.



Translation Table

Private	External
172.18.3.1	25.8.2.10

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→ In this strategy, communication must always be initiated by the private network.

Network Address Translation (NAT) Types:

There are 3 ways to configure NAT.

1) Static NAT: In this, a single unregistered (private) IP address is mapped with a legally registered (public) IP address. i.e. one-to-one mapping b/w local and global addresses. This is generally used for web hosting. These are not used in organizations as there are many devices that will need Internet access and to provide internet access, a public IP address is needed.

eg: 3000 devices ⟶ 3000 public IP addresses

2) Dynamic NAT: In this type of NAT, an unregistered IP address is translated into a registered (public) IP address from a pool of public IP addresses. If the IP address pool is not free, then the packet will be dropped as only fixed no. of private IP address can be translated to public addresses.

pool

[2 IP] ⟹ only 2 private addresses can be translated

→ If 3rd private address wants to access to internet the packet will be dropped therefore many private addresses are mapped to pool of public IP addresses.

→ NAT is used when the number of users who want to access internet is fixed.

→ This is also costly as the organization has to buy many global IP addresses to make pool.

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Port address Translation (PAT): This is also know as NAT overload. In this many local (private) IP address can be translated to a single registered IP address. Port no's used to distinguish traffic (i.e.) which traffic belongs to which IP address. This is mostly used as it is cost-effective as thousands of users can be connected to the Internet by using only one real global (public) IP address.

Why mask port numbers:

Suppose, in a network, two hosts A and B are connected. Now both of them request for same destination, on the same port number say 1000, on the host side, at the same time. If NAT does only translation of IP addresses, then when their packets ~~at~~ will arrive at the NAT, both of their IP addresses would be masked by the public address of the network and sent to the destination. Destination will send replies to the public IP address of the router. Thus, receiving reply, it will be unclear to NAT as to which ~~to~~ reply belongs to which host (bcoz sourceport numbers for both A and B are the same). Hence to avoid such a problem, NAT masks the source port number as well and makes an entry in the NAT table.

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